

PRAKTIKUM SISTEM OPERASI

MODUL X

SHELL



Disusun Oleh :

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JURNAL

1. Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall. Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

a. Pada prototypes.h chname diubah menjadi:

```
extern status ascdade(uint32, char *);
-----
/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);

/* in file clkupdate.S */
extern uint32 clkcount(void);
```

b. Pada chname.c fungsi diubah dari:

```
/* cname.c - chname */

// Nama: ALYA RABANI
// NIM: 2311104076

#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
syscall chname(
    pid32      pid,
    char *     newname
)
```

menjadi

```
/* chame.c - cname */

// Nama: ALYA RABANI
// NIM: 2311104076

#include <xinu.h>

/*-----
 * cname - Dummy syscall
 *-----
 */
pri16 cname(
    pid32      pid,
    char *     *newname,
    uint32     namelen
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil

```

*/
pri16 chname(
    pid32      pid,
    char       *newname,
    uint32     namelen
)
{
    intmask mask;
    struct procent *prptr;
    uint32 i;

    mask = disable();

    /* Validasi PID */
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];

    /* Nama baru */
    for (i = 0; i < namelen && i < PNMLEN-1; i++) {
        prptr->prname[i] = newname[i];
    }

    /* Null terminator */
    prptr->prname[i] = '\0';

    restore(mask);
    return OK;
}

```

2. Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

a. Edit shell.c

```

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/*****
 * Table of Xinu shell commands and the function associated with each
 *****/
const struct cmdent cmdtab[] = {
    {"argecho", xsh_argecho},
    {"arp", xsh_arp},
    {"cat", xsh_cat},
    {"clear", xsh_clear},
    {"date", xsh_date},
    {"devdump", xsh_devdump},
    {"echo", xsh_echo},
    {"help", xsh_help},
    {"ls", xsh_ls},
    {"kill", xsh_kill},
    {"memdump", xsh_memdump},
    {"memstat", xsh_memstat},
    {"ns", xsh_ns},
    {"netinfo", xsh_netinfo},
    {"ping", xsh_ping},
    {"ps", xsh_ps},
    {"sleep", xsh_sleep},
    {"tee", xsh_tee},
    {"udp", xsh_udpdump},
    {"udpecho", xsh_udpecho},
    {"udpserver", xsh_udpserver},
    {"uptime", xsh_uptime},
    {"namecmd", xsh_namecmd},
    {"mycmd", xsh_mycmd},
    {"?", xsh_help}
}

```

b. Edit shprototypes.h

```

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);

```

c. Buat file baru:

```

/* xsh_namecmd.c */
// Nama: ALYA RABANI
// NIM: 2311104076

#include <xinu.h>
#include <stdio.h>
#include <string.h>

shellcmd xsh_namecmd(int32 nargs, char *args[])
{
    printf("My new command\n");

    /* pid 3 = neti, ubah nama proses menjadi hello */
    chname(2, "hello", 5);

    return 0;
}

```

3. Test hasilnya:
 - a. Masuk ke terminal xinu
 - b. Jalankan perintah ps

```

XINU
-----2311104076-----
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size  Msg  Content
-----
0 pmnull      ready  0    0 0x005F0FFC 0x00FFFECC 8192 1 3
1 ipout       wait   500  0 0x005F0FFC 0x005F0F30 8192 0 0
2 netin       wait   500  0 0x005F0FFC 0x005F0ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005F7F34 65536 0 0
5 shell       recv   50   4 0x005F79AC 0x005F79AC 8192 0 0
6 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $

```

- c. Jalankan perintah namecmd

```

XINU
-----2311104076-----
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size  Msg  Content
-----
0 pmnull      ready  0    0 0x005F0FFC 0x00FFFECC 8192 1 3
1 ipout       wait   500  0 0x005F0FFC 0x005F0F30 8192 0 0
2 netin       wait   500  0 0x005F0FFC 0x005F0ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005F7F34 65536 0 0
5 shell       recv   50   4 0x005F79AC 0x005F79AC 8192 0 0
6 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $

```

- d. Jalankan perintah ps

```
Welcome to Xinu!

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
 0 prnull                ready  0    0 0x005FDFFC 0x00FFFECC    8192  1    3
 1 ipout                  wait  500  0 0x005FBFFC 0x005FBF30    8192  0    0
 2 netin                  wait  500  0 0x005F9FFC 0x005F9ED0    8192  0    0
 4 Main process           recv  20   3 0x005E7FFC 0x005E7F34   65536  0    0
 5 shell                  recv  50   4 0x005F7FFC 0x005F79AC    8192  0    0
 6 ps                     curr  20   5 0x005F5FFC 0x005F5FC4    8192  0    0

xsh $ namecmd
My new command
xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
 0 prnull                ready  0    0 0x005FDFFC 0x00FFFE08    8192  1    3
 1 ipout                  wait  500  0 0x005FBFFC 0x005FBF30    8192  0    0
 2 hello                  wait  500  0 0x005F9FFC 0x005F9ED0    8192  0    0
 4 Main process           recv  20   3 0x005E7FFC 0x005E7F34   65536  0    0
 5 shell                  recv  50   4 0x005F7FFC 0x005F79AC    8192  0    7
 8 ps                     curr  20   5 0x005F5FFC 0x005F5FC4    8192  0    0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- e. Lihat nama proses telah berubah

Nama proses berubah dari *netin* menjadi *hello* karena syscall `chname()` berhasil dijalankan melalui command `namecmd`.